Software Requirements Specification

Agriculture

Structured technical document aligned for formal review, sign-off, and handover.

Document ID	**Version**	**Date**	**Author**
SRS-20260418-141308	1.0	2026-04-18	Ali

Req2Design IEEE 830-1998 aligned Generated draft for expert review

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Revision History

Name	Date	Reason For Changes	Version
Ali	2026-04-18	Initial generated draft for review.	1.0

No other changes to the provided structure or format are permitted.

1.

1: Introduction

1.1: Purpose

Purpose: To define the software requirements for an agricultural management system designed to assist farmers, field officers, and administrators in optimizing crop yields through data-driven decision-making.

1.2: Scope

Scope: This Software Requirements Specification (SRS) covers the functionalities required for registering land profiles, managing crop cycles, providing localized advisories, tracking farmer visits, and monitoring advisory adoption rates.

1.3: Definitions/Acronyms

- **FARMER:** A user who owns or manages farmland and seeks advice on farming practices.
- **FIELD OFFICER:** An agent responsible for visiting farms, recording observations, and recommending actions based on local conditions.
- **ADMINISTRATOR:** A user with oversight responsibilities over multiple farmers and field officers.
- **LAND PROFILE:** Information about a specific piece of farmland including its location, size, type, and historical data.
- **CROP CYCLE:** The sequence of planting, growing, harvesting, and post-harvest activities associated with a particular crop.
- **LOCALIZED ADVISORY:** Customized recommendations tailored to the specific environmental and socio-economic conditions of a farm.
- **WEATHER FORECAST:** Predictions of meteorological conditions relevant to agriculture.
- **SOIL DATA:** Information about the physical and chemical properties of soil, such as pH levels, nutrient content, and moisture retention capacity.
- **SEASONAL ALERTS:** Notifications regarding critical times during the year when certain actions may have significant impact on crop health or yield.
- **RECOMMENDATION HISTORY:** A record of all past suggestions made to a farmer by field officers or the system itself.

- **VILLAGE CLUSTER:** A geographical grouping of farms within a defined area used for organizing field officer visits.
- **LOW-COMMUNITY:** Areas with limited internet connectivity suitable for offline data collection followed by periodic synchronization.

2.

2: Overall Description

Perspective: The perspective is that of a software application aimed at supporting various stakeholders involved in agricultural management.

Functions: The system will provide functionalities for registration, advisory generation, visit logging, and administrative reporting.

Users: Farmers, field officers, and administrators.

2.4: Constraints

Constraints: The system must operate reliably even under poor network conditions, especially in rural settings.

2.5: Assumptions/Dependencies

Assumptions/Dependencies: The system assumes access to external services for weather forecasting and soil analysis. It also depends on accurate input from users to generate meaningful outputs.

3.

3: Specific Requirements

External Interfaces:

- Interface with external weather service providers for real-time weather updates.
- Interface with soil testing laboratories for soil data acquisition.
- Interface with mobile devices for offline data entry and synchronization.
- Interface with third-party mapping services for geolocation-based services.

4: FUNCTIONAL REQUIREMENTS

FR-01

- Land Profile Registration

Input: Farmer inputs details about their land profile including geographic coordinates, size, type, and previous crop history.

Processing: The system validates the entered information against predefined criteria and stores it securely.

Output: Confirmation message indicating successful registration along with any necessary corrections.

Priority: High

FR-02

- Crop Cycle Management

Input: Farmer selects crops they intend to plant and enters expected dates for key events like sowing, flowering, and harvest.

Processing: The system generates a timeline of tasks and milestones related to the selected crop cycle.

Output: Displayed calendar view showing planned and actual progress.

Priority: Medium

FR-03

- Localized Advisory Generation

Input: Farmer submits current conditions and needs assistance with irrigation, fertilization, or pest control.

Processing: The system analyzes available data including weather forecasts, soil reports, and historical records to formulate personalized advice.

Output: Detailed report outlining recommended actions and rationale behind them.

Priority: High

FR-04

- Field Officer Visit Logs

Input: Field officer records observations and recommendations during visits to different villages.

Processing: The system associates these entries with corresponding land profiles and maintains a log of interactions.

Output: Summary report highlighting trends and insights derived from collected data.

Priority: Low

FR-05

- Administrator Monitoring Tools

Input: Administrator views aggregated statistics on advisory usage patterns, crop success rates, and pending queries.

Processing: The system compiles and presents comprehensive analytics based on stored data.

Output: Interactive dashboard displaying key metrics and visualizations.

Priority: Medium

5: NON-FUNCTIONAL REQUIREMENTS

Usability:

- The interface shall be intuitive and easy to navigate for users across different age groups and literacy levels.
- Data entry forms shall minimize errors through clear labeling and validation checks.

Reliability:

- The system must ensure high uptime and quick recovery from failures.
- Offline functionality shall allow reliable operation in remote locations.

Performance:

- Response time for critical operations such as generating advisories shall not exceed five seconds.
- Large datasets shall load within ten seconds.

Portability:

- The application shall run seamlessly on both desktop computers and mobile devices.
- Support for multiple languages ensures accessibility worldwide.

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